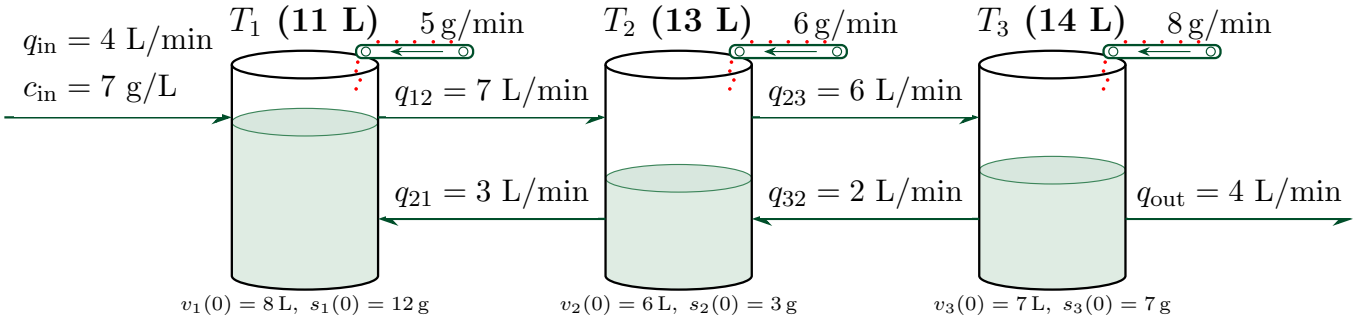
1

MA3521 — Exam One Key

Name:



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 12 \\ 3 \\ 7 \end{pmatrix}$ g $\quad 0 < t <$ $\boxed{\infty}$ min

$\mathbf{v}(t) =$ $\boxed{\begin{pmatrix} 8 \\ 6 \\ 7 \end{pmatrix}}$ L $\qquad \mathbf{b} =$ $\boxed{\begin{pmatrix} 33 \\ 6 \\ 8 \end{pmatrix}}$ g/min

(b) $\boldsymbol{A} =$ $\boxed{\begin{pmatrix} -7/8 & 3/6 & 0 \\ 7/8 & -9/6 & 2/7 \\ 0 & 6/6 & -6/7 \end{pmatrix}}$ min^{-1}

Brine and dry salt inputs (g/min)
$b_1 = 4(7) + 5 = 33, \quad b_2 = 6, \quad b_3 = 8$

(c) A UC form is $\mathbf{s}_p(t) =$ $\boxed{\mathbf{a}}$. $\boldsymbol{A}\mathbf{a} = -\mathbf{b}$ Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} =$ $\boxed{\left[\begin{array}{ccc|c} -7/8 & 3/6 & 0 & -33 \\ 7/8 & -9/6 & 2/7 & -6 \\ 0 & 6/6 & -6/7 & -8 \end{array}\right]}$

Net liquid flow by tank (L/min)
$4 + 3 - 7 = 0, \quad 7 + 2 - 3 - 6 = 0$
$6 - 2 - 4 = 0$

(a) ☑ Linear ☐ Separable $\quad ty' - 3y = t^4 e^{2t} \qquad y(1) = 1$

Show a suitable intermediate step

$$(t^{-3}y)' = e^{2t}$$

$$y' - \frac{3}{t}y = t^3 e^{2t}, \quad \mu(t) = e^{-3\ln t} = t^{-3}$$
$$t^{-3}y = \tfrac{1}{2}e^{2t} + C, \quad C = 1 - \tfrac{1}{2}e^2$$

Show a fully integrated answer

$$t^{-3}y = 1 + \tfrac{1}{2}(e^{2t} - e^2)$$

(b) ☐ Linear ☑ Separable $\quad y' = 3t^2\cos(t^3)(1+y)^2 \qquad y(0) = 1$

Show a suitable intermediate step

$$\int \frac{dy}{(1+y)^2} = \int 3t^2\cos(t^3)\,dt$$

$$u = t^3, \quad du = 3t^2\,dt$$
$$-\frac{1}{1+y} = \sin(t^3) + C, \quad y(0) = 1 \implies C = -\tfrac{1}{2}$$

Show a fully integrated answer

$$-\frac{1}{1+y} = \sin(t^3) - \tfrac{1}{2}$$

3

$$\mathbf{y}' = \boldsymbol{A}\mathbf{y} + e^{3t}\begin{pmatrix}2\\-1\end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix}1 & -2\\ 8 & 1\end{pmatrix} \quad \mathbf{y}(0) = \begin{pmatrix}1\\3\end{pmatrix}$$

Eigenpairs $1+4i$, $\begin{pmatrix}1\\-2i\end{pmatrix}$ $\quad 1-4i$, $\begin{pmatrix}1\\2i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\boxed{c_1 e^t\begin{pmatrix}\cos(4t)\\2\sin(4t)\end{pmatrix} + c_2 e^t\begin{pmatrix}\sin(4t)\\-2\cos(4t)\end{pmatrix} \quad \text{or} \quad e^t\begin{pmatrix}\cos(4t) & \sin(4t)\\ 2\sin(4t) & -2\cos(4t)\end{pmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix}}$

$$\mathbf{v} = \begin{pmatrix}1\\0\end{pmatrix} + i\begin{pmatrix}0\\-2\end{pmatrix}$$
$$e^{(1+4i)t} = e^t\cos(4t) + ie^t\sin(4t)$$
$$\text{re}\big(\mathbf{v}e^{(1+4i)t}\big) = e^t\begin{pmatrix}\cos(4t)\\2\sin(4t)\end{pmatrix}$$
$$\text{im}\big(\mathbf{v}e^{(1+4i)t}\big) = e^t\begin{pmatrix}\sin(4t)\\-2\cos(4t)\end{pmatrix}$$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$ $\boxed{e^{3t}\mathbf{a}}$

$$\mathbf{y}_p' = 3e^{3t}\mathbf{a}$$
$$\boldsymbol{A}\mathbf{y}_p + e^{3t}\begin{pmatrix}2\\-1\end{pmatrix}$$
$$= e^{3t}\left(\boldsymbol{A}\mathbf{a} + \begin{pmatrix}2\\-1\end{pmatrix}\right)$$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$ $\boxed{\left[\begin{array}{cc|c} 2 & 2 & 2\\ -8 & 2 & -1\end{array}\right]}$

$$3\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix}3 & 0\\ 0 & 3\end{pmatrix} - \begin{pmatrix}1 & -2\\ 8 & 1\end{pmatrix}$$
$$= \begin{pmatrix}2 & 2\\ -8 & 2\end{pmatrix}$$

(d) To satisfy the IC you would row reduce

$\boldsymbol{M}_{\text{IC}} =$ $\boxed{\left[\begin{array}{cc|c} 1 & 0 & 1-a_1\\ 0 & -2 & 3-a_2\end{array}\right]}$

$$\mathbf{y}_c(0) = \begin{pmatrix}c_1\\-2c_2\end{pmatrix}, \quad \mathbf{y}_p(0) = \begin{pmatrix}a_1\\a_2\end{pmatrix}$$
$$\mathbf{y}_c(0) + \mathbf{y}_p(0) = \begin{pmatrix}1\\3\end{pmatrix}$$

$$\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \sin(3t)\begin{pmatrix}2\\1\end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix}2 & 1\\0 & -1\end{pmatrix}$$

Eigenpairs 2, $\begin{pmatrix}1\\0\end{pmatrix}$ $\quad -1,\ \begin{pmatrix}1\\-3\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\boxed{c_1 e^{2t}\begin{pmatrix}1\\0\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}1\\-3\end{pmatrix} \quad \text{or} \quad \begin{pmatrix}e^{2t} & e^{-t}\\0 & -3e^{-t}\end{pmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix}}$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$ $\boxed{\mathbf{a}\cos(3t) + \mathbf{b}\sin(3t)}$

$$\begin{aligned}\mathbf{y}_p' &= 3\mathbf{b}\cos(3t) - 3\mathbf{a}\sin(3t)\\ \boldsymbol{A}\mathbf{y}_p + \begin{pmatrix}2\\1\end{pmatrix}&\sin(3t)\\ &= (\boldsymbol{A}\mathbf{a})\cos(3t)\\ &+ \left(\boldsymbol{A}\mathbf{b} + \begin{pmatrix}2\\1\end{pmatrix}\right)\sin(3t)\end{aligned}$$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$ $\boxed{\left[\begin{array}{cc|c} -\boldsymbol{A} & 3\boldsymbol{I} & \begin{pmatrix}0\\0\end{pmatrix}\\ -3\boldsymbol{I} & -\boldsymbol{A} & \begin{pmatrix}2\\1\end{pmatrix}\end{array}\right]}$

$$-\boldsymbol{A}\mathbf{a} + 3\boldsymbol{I}\mathbf{b} = \begin{pmatrix}0\\0\end{pmatrix} \quad (\cos)$$

$$-3\boldsymbol{I}\mathbf{a} - \boldsymbol{A}\mathbf{b} = \begin{pmatrix}2\\1\end{pmatrix} \quad (\sin)$$

$$-\boldsymbol{A} = \begin{pmatrix}-2 & -1\\0 & 1\end{pmatrix}$$

Columns: a_1, a_2, b_1, b_2